Seminar in Laboratory of Visual Intelligence and Pattern Analysis (VIPA)

Recent Advances in Image Reconstruction

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Outline

- Image Reconstruction
  - Introduction
  - Recent Advances
- Discussing Two Related NIPS Papers
  - #1. Generating Images with Perceptual Similarity Metrics
  - #2. Synthesizing the Preferred Inputs for Neurons in Neural Networks **(Application of #1)**
- Personal Comments and Ideas

Image Reconstruction: Introduction

- Image Reconstruction:
  - Roughly estimate the **original image** from **image representations**.
  - A process of image generation.

INPUT

Image representation (y)

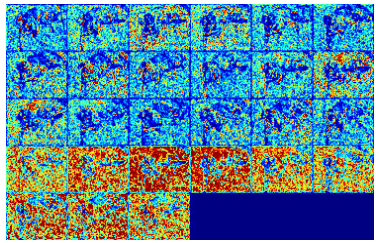

Image Reconstruction Algorithm

OUTPUT

Original image (x) corresponding to y

Image Reconstruction: Introduction

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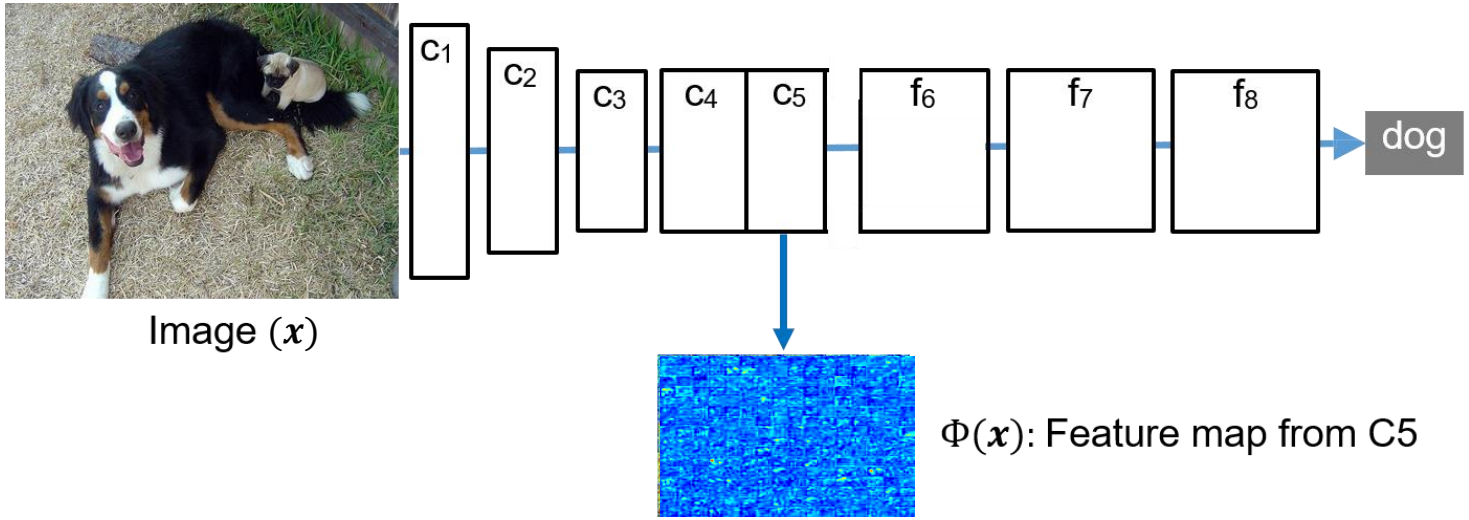
  - e.g.



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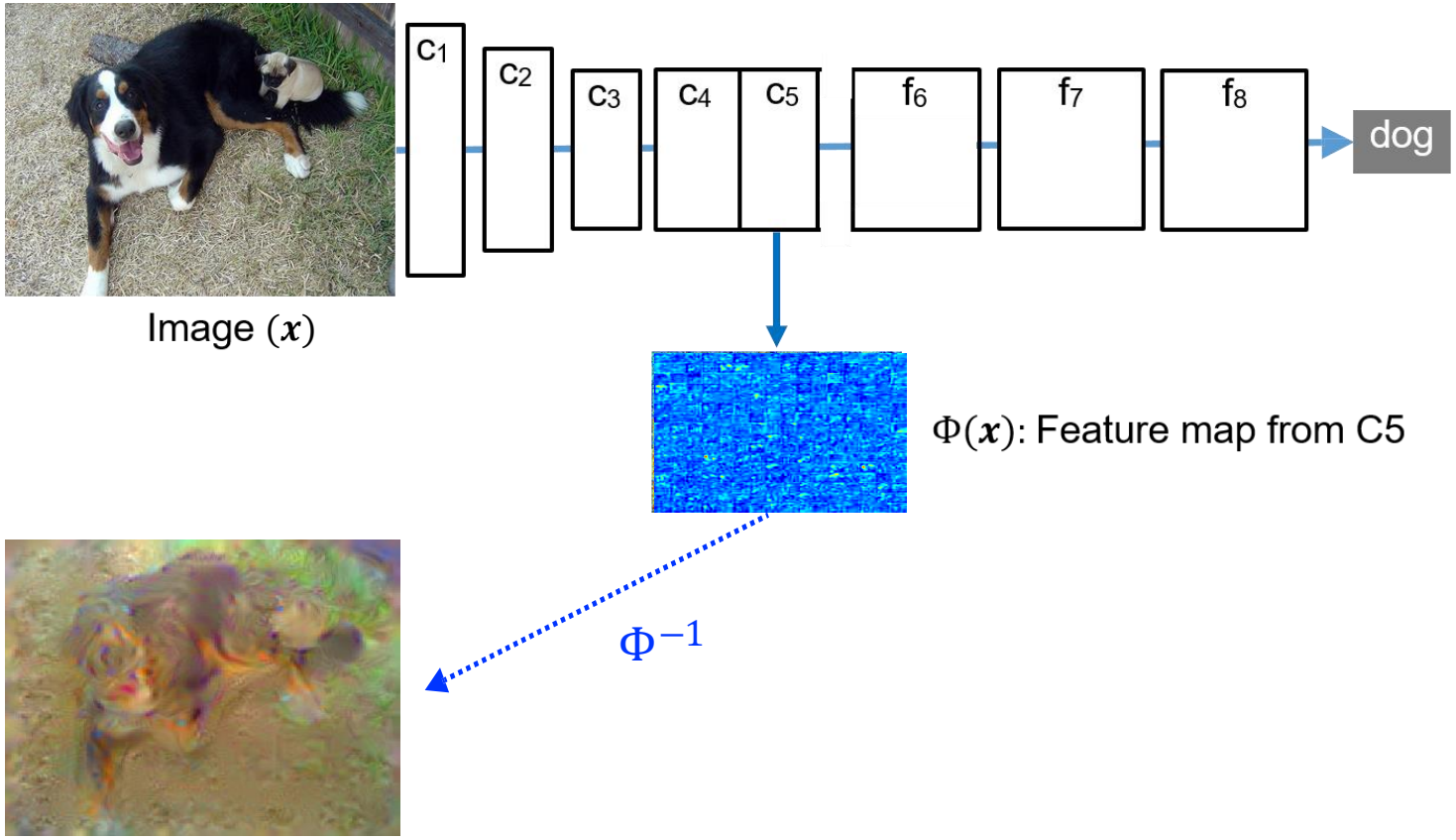

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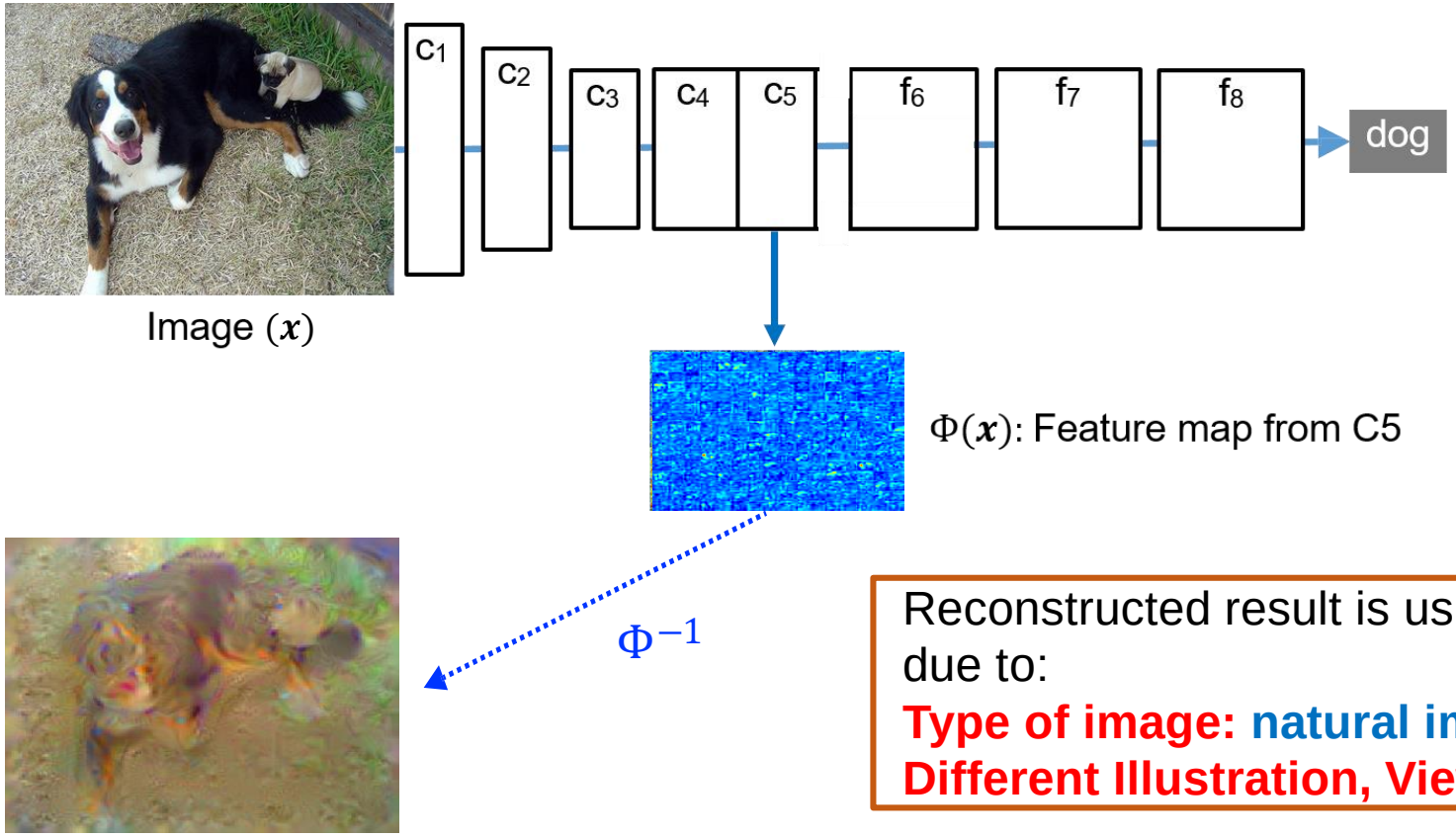
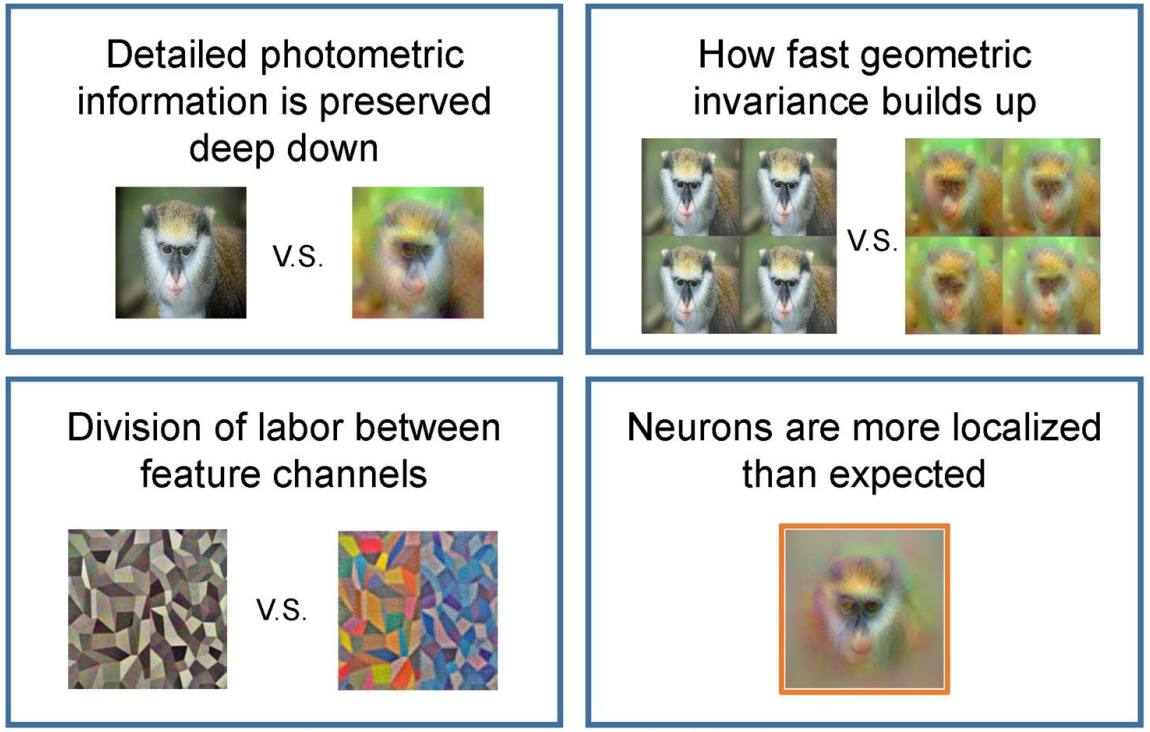


Reconstructed result is usually **not unique**,
due to:
Type of image: natural image, non-natural image;
Different Illustration, Viewpoint, etc.

Image Reconstruction: Introduction

- Image Reconstruction:
  - **Application**
    - #1. **Understanding** and **Visualizing** CNN



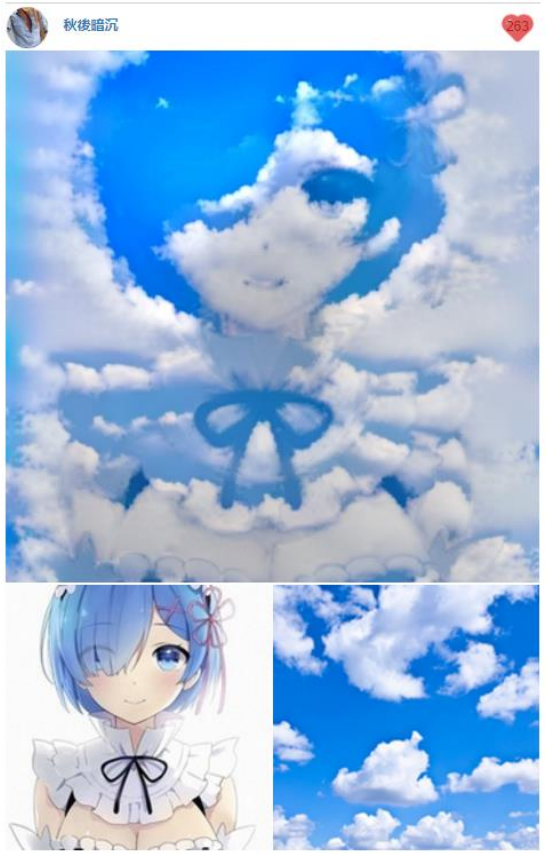
Understanding Deep Image Representations by Inverting Them 47

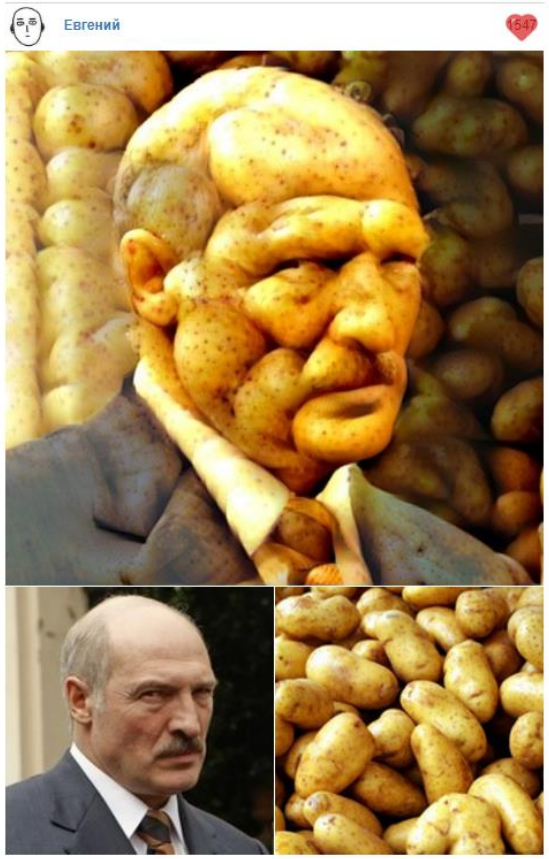
- Visualizing **what CNN learns**
  - Judge architecture
  - See if it is overfitting
- Visualizing the **data gradient**
  - Improve training
- Fooling CNN (Adversarial Examples)

1. http://cs231n.github.io/understanding-cnn/

Image Reconstruction: Introduction

- Image Reconstruction:
  - **Application**
    - #2. **Creating art** by synthesizing different representations





- Neural Style
- Feature Interpolation

2. Images from http://www.ostagram.ru/

Image Reconstruction: Introduction

- Image Reconstruction:
  - **Application**
    - #3. **Generating** similar natural images with the same **semantic content**

3. Images are generated using algorithms in NIPS 2016 paper “Synthesizing the preferred inputs for neurons in neural networks via deep generator networks”

Image Reconstruction: Introduction

- Image Reconstruction:
  - **Potential Application**
    - #4. Reconstructing images from **measured brain activity**
      - Personally, the process of image reconstruction is **similar** to the process of **imaging something in human's brain (abstract -> concrete)**
      - Visualize what people imagine in the brain

Image Reconstruction: Recent Advances

- ① Understanding Deep Image Representations by Inverting Them
  - Pioneering work (has introduced in my last, last seminar)
  - CVPR 2015 oral **(Based on image iteration)**
  - Main idea: **hand-designed natural image prior** to reconstruct only natural image

$$E(x) = \|\varphi(x) - \varphi_0\|^2 \boxed{+ \lambda R(x)}$$

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- ② Inverting Visual Representations with Convolutional Networks
  - CVPR 2016 poster **(Based on model iteration)**
  - Main idea: “Rudely” **train a specific network** to reconstruct feature map from each layer.
  - Results are not that satisfying.

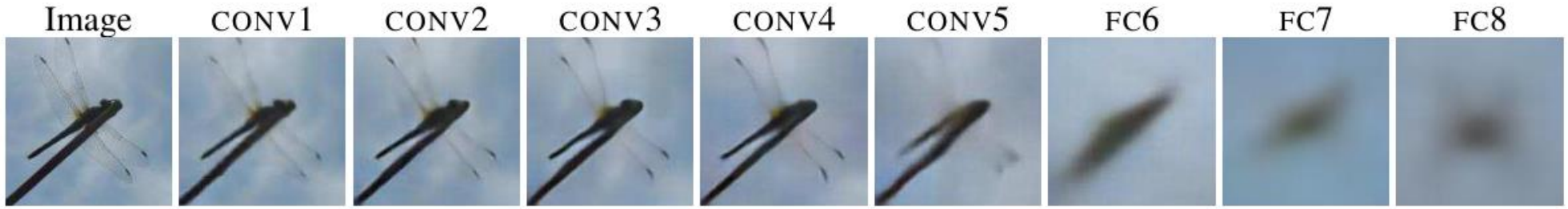
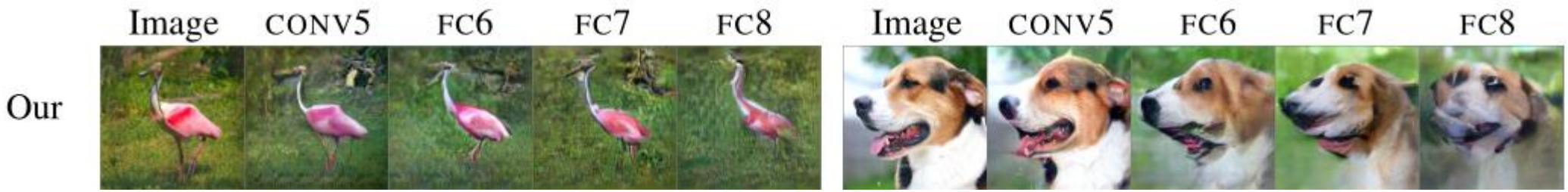


Image Reconstruction: Recent Advances

- ③ Generating Images with Perceptual Similarity Metrics based on Deep Networks
  - NIPS 2016 **(Based on model iteration)**
  - Main idea: **Exploit "D" in GAN** to distinguish natural images during image reconstruction
  - Big improvement ☺



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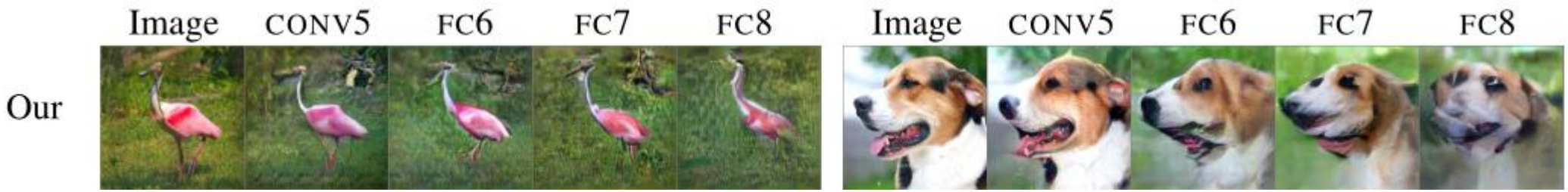

- ④ Synthesizing the Preferred Inputs for Neurons in Neural Networks via Deep Generator Networks
  - NIPS 2016 **(Extension and application of image reconstruction)**
  - Main idea:
    - Exploit **trained "G" in ③** as the natural image prior (improvement over hand-designed priors).
    - Optimize the **code** (input of FC layer) **instead of image** to reconstruct the image.

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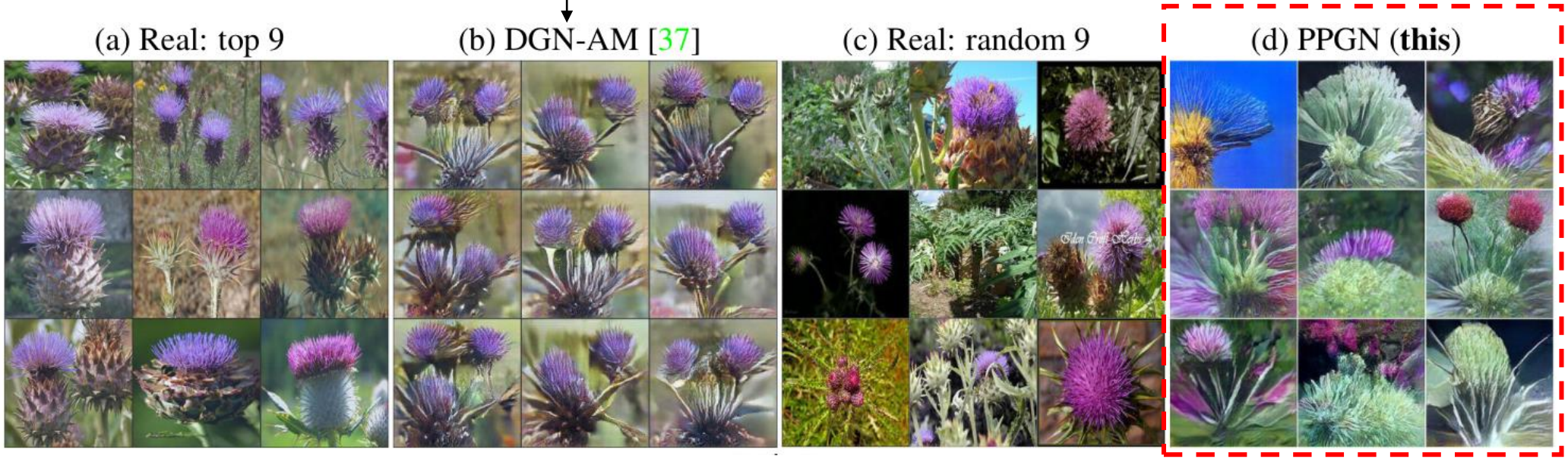
TODAY'S FOCUS!

Image Reconstruction: Recent Advances

- ⑤ Plug & Play Generative Networks: Conditional Iterative Generation of Images in Latent Space
  - CVPR 2017 spotlight
  - **Yoshua Bengio**'s paper
  - Totally 36 pages including appendix with all kinds of mathematical proof
  - Explosive improvement over ④
  - Maybe next time …



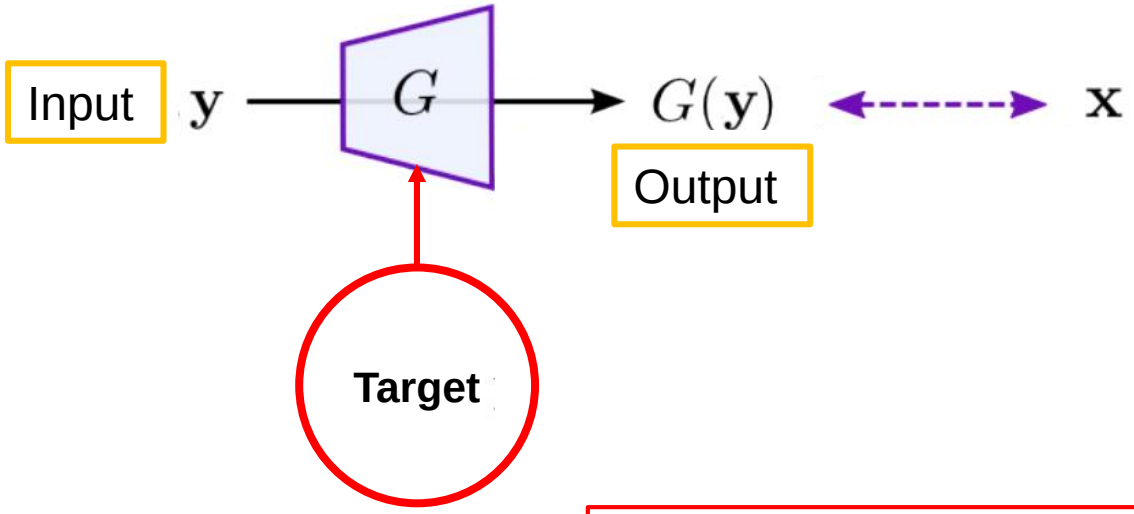
Generating Images with Perceptual Similarity Metrics based on Deep Networks

NIPS 2016

Alexey Dosovitskiy and Thomas Brox

Introduction

- Image reconstruction from high-level abstract representation
  - Input: representation $\boldsymbol{y}$
  - Output: reconstructed image $\boldsymbol{G(y)}$
  - Target: For representation in **each layer** to be constructed, train a **layer-specific Generator** to reconstruct images.



Architecture of proposed model

Introduction

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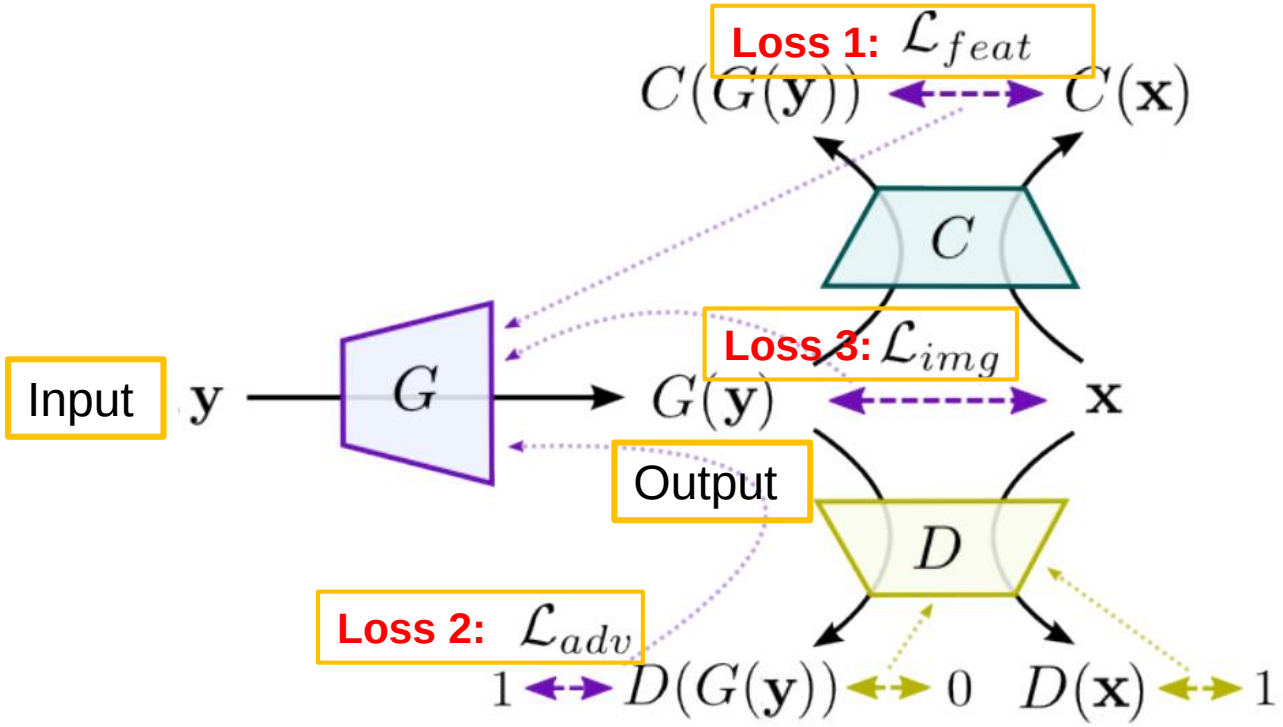


Architecture of proposed model

Exploit GAN to train

Proposed Method

- Symbol definition

(Constant) • Input representation y

(Variable) • Generator remaining to be trained G

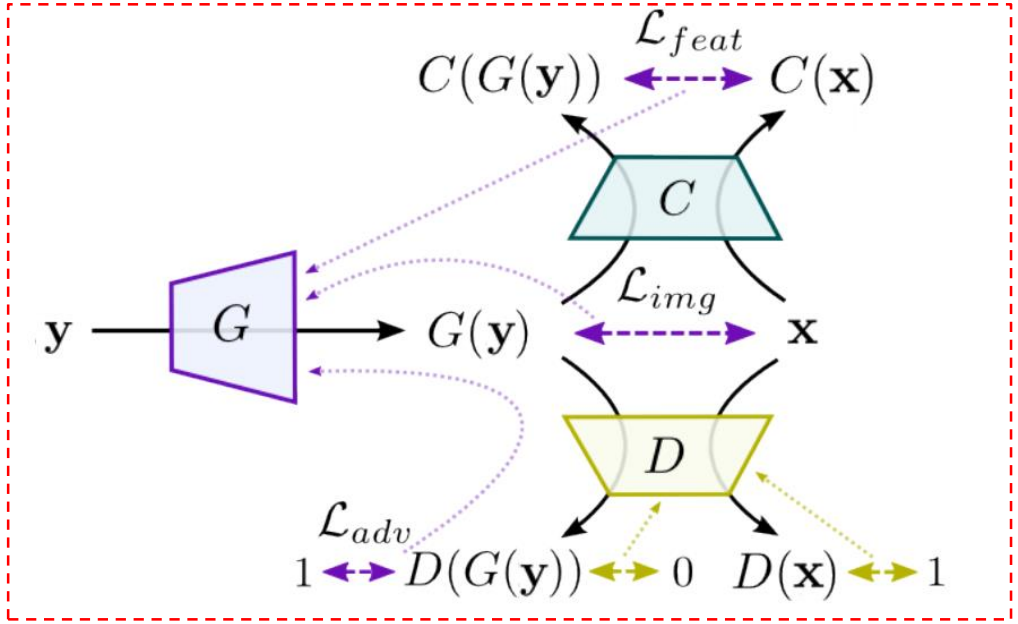
• Output reconstructed image $G(y)$

(Constant) • Original image x

(Variable) • Discriminator D (discriminate natural images)

(Constant) • Comparator C (**A layer in a CNN** that we want to reconstruct from)

• Loss L



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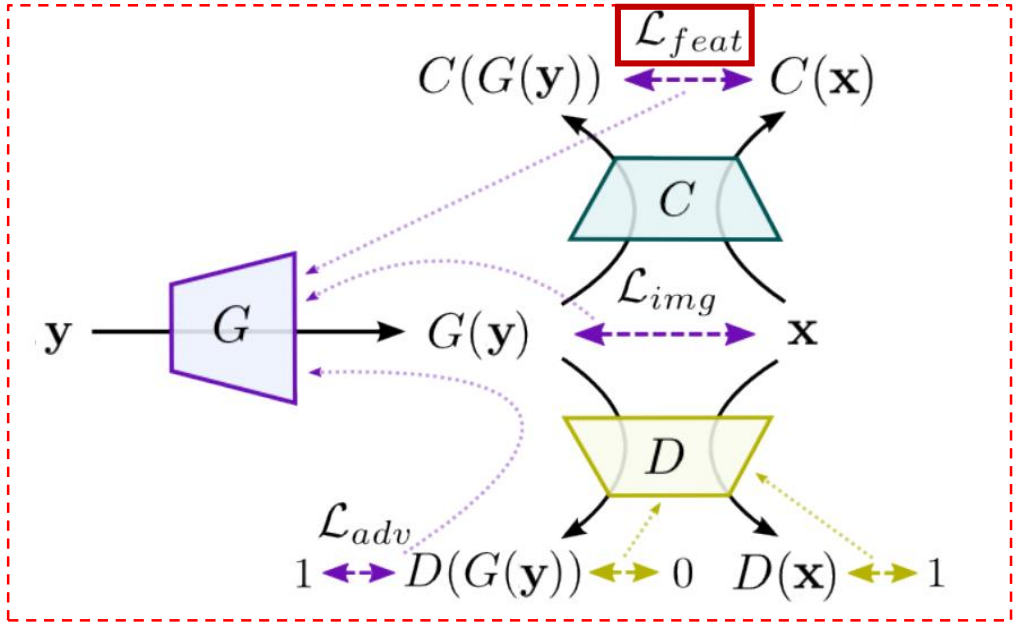

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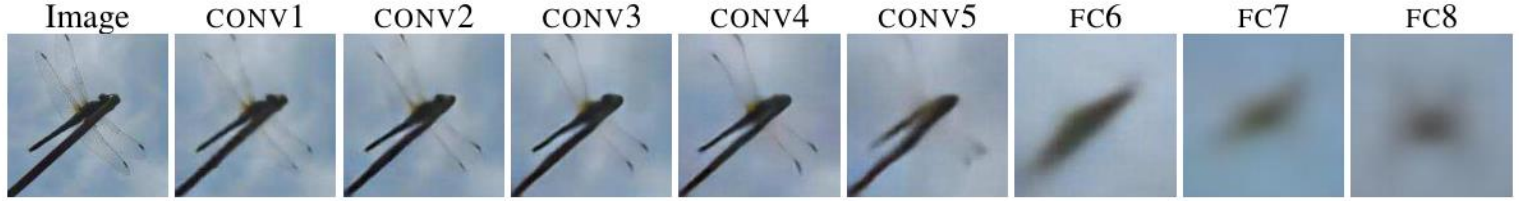
  - Loss L

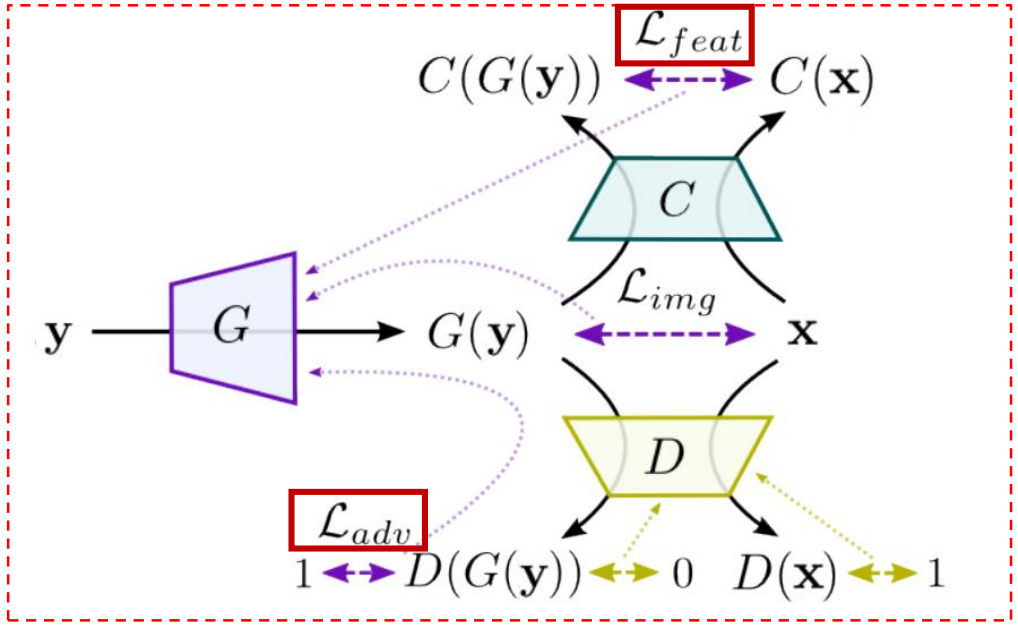
- $L = \sum_i \left\| C\left(G(y_i)\right) - C(x_i) \right\|_2^2 \longrightarrow \boxed{L_{feat}\text{: loss in feature space, or perceptual loss}}$
  - Force the representation of generated reconstructed image to be equal to representation of x
  - **Only** use this loss is exactly the algorithm in ② CVPR 2016 poster paper (refer to Slide p12 if you forget)
  - **Blurry results for FC layer:**



Proposed Method

$\boxed{L_{feat}}$

- $L = L_{adv} + \lambda_{feat} \sum_i \|G(y_i) - x_i\|_2^2$
  - Add another **adversarial loss** to learn natural image prior



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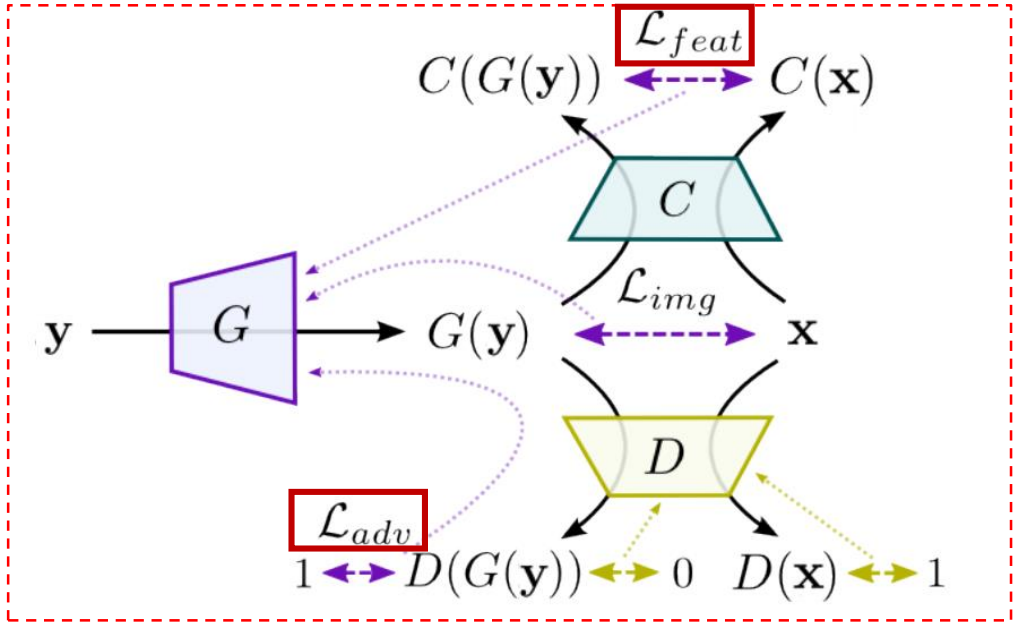
- $L_{adv} = -\sum_i \log D(G(y_i))$
  - Common loss for generator in GAN

- $\boldsymbol{D}$ is optimize using:
  - $L_{discr} = -\sum_i \log\big(D(x_i)\big) + \log(1 - D(G(y_i)))$



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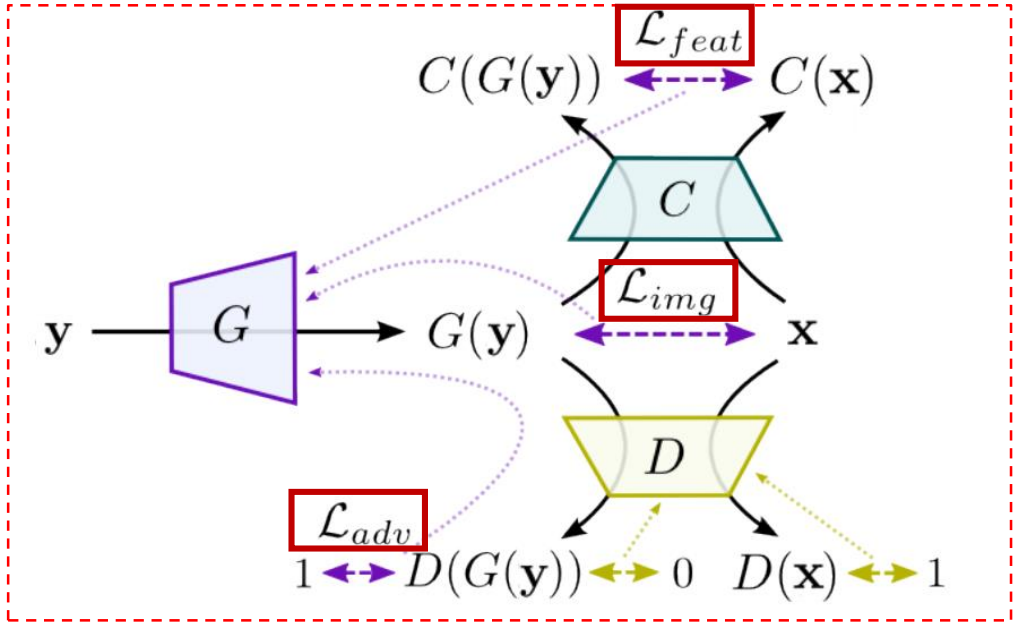

- One remaining problem: parameter tuning is a little bother
  - Add $\mathbf{L_{img}}$（**loss in image space**）to make parameter tuning easier
  - $L_{img} = \sum_i \|G(y_i) - x_i\|_2^2$ (pixel value subtraction)

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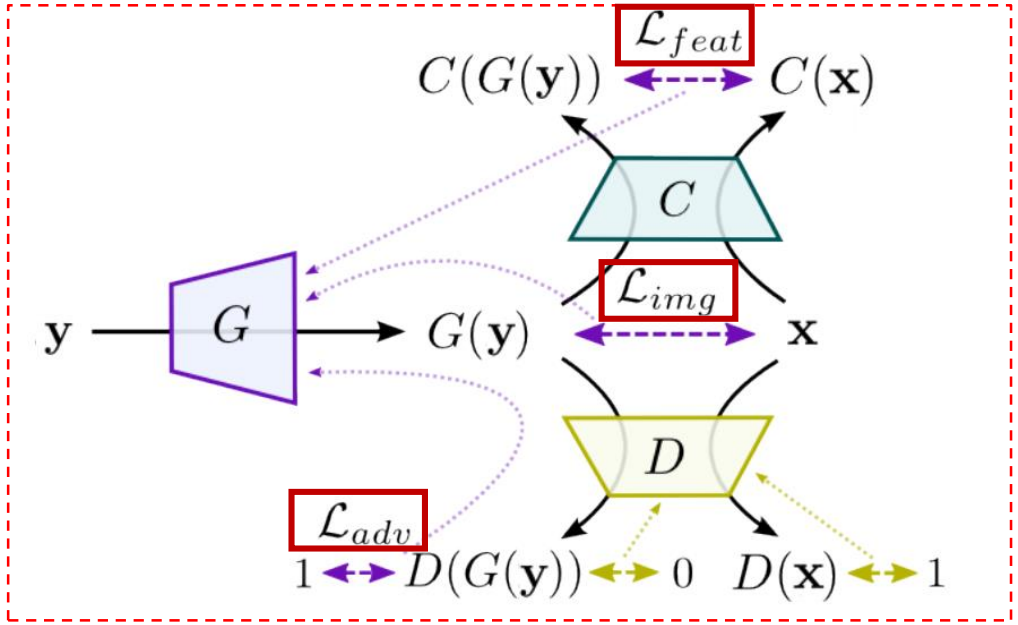
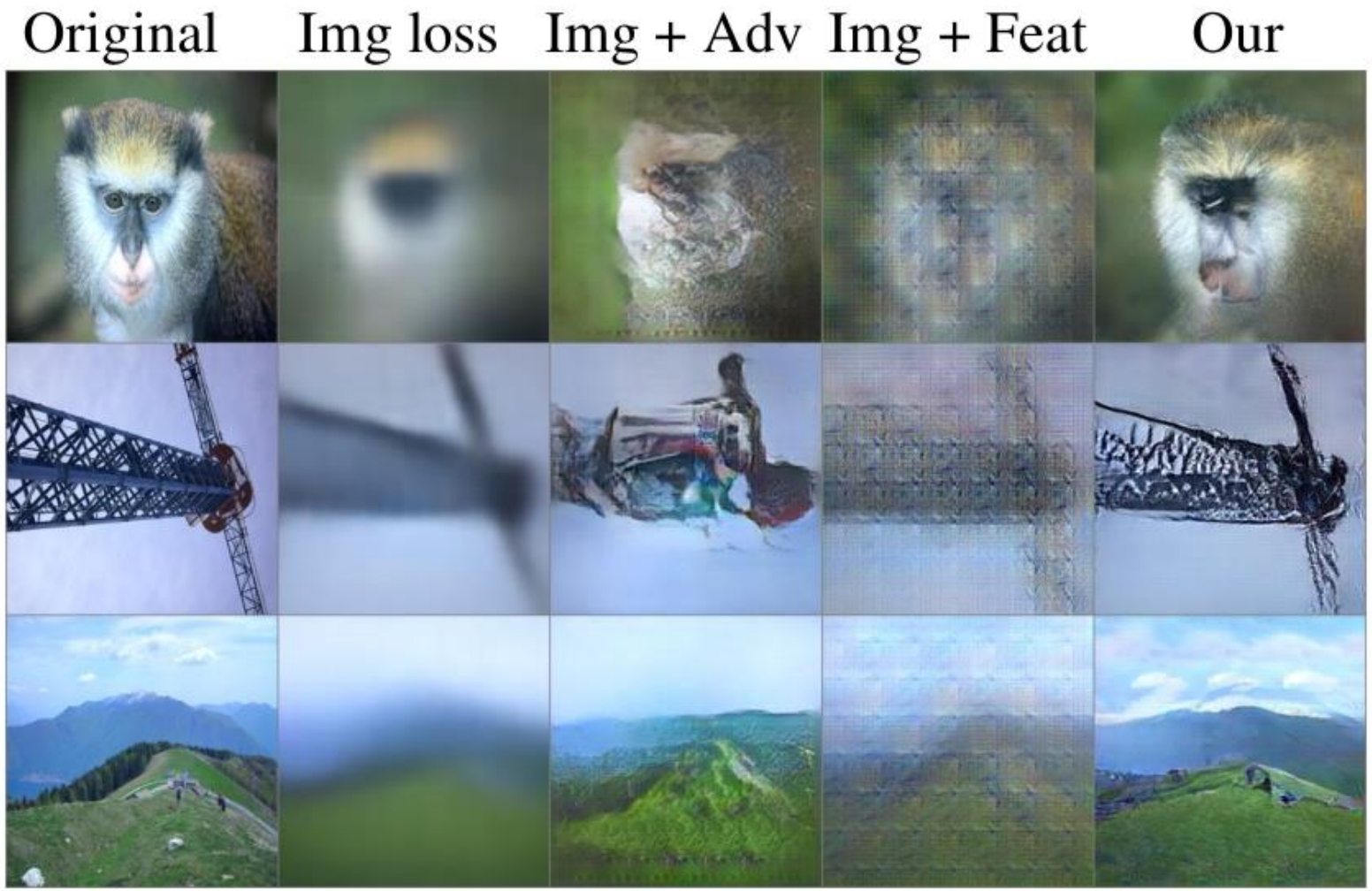


- Overall, we get our final loss:

$$\boldsymbol{L_{total} = \lambda_{feat} L_{feat} + \lambda_{adv} L_{adv} + \lambda_{img} L_{img}}$$

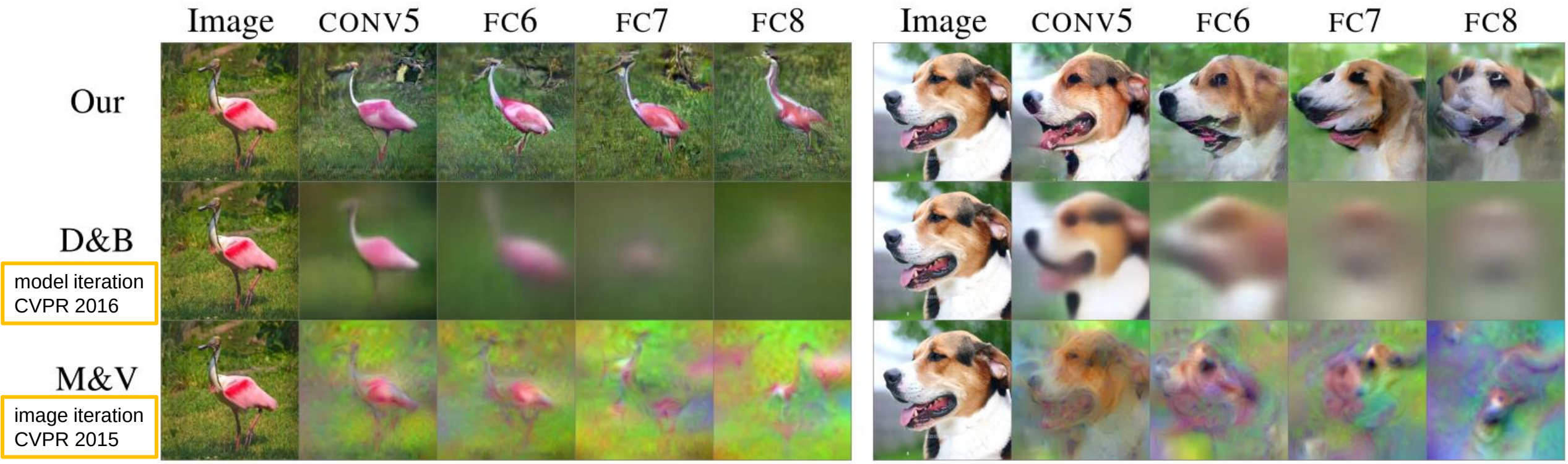
Results

- Using **all three losses** produces the best results.



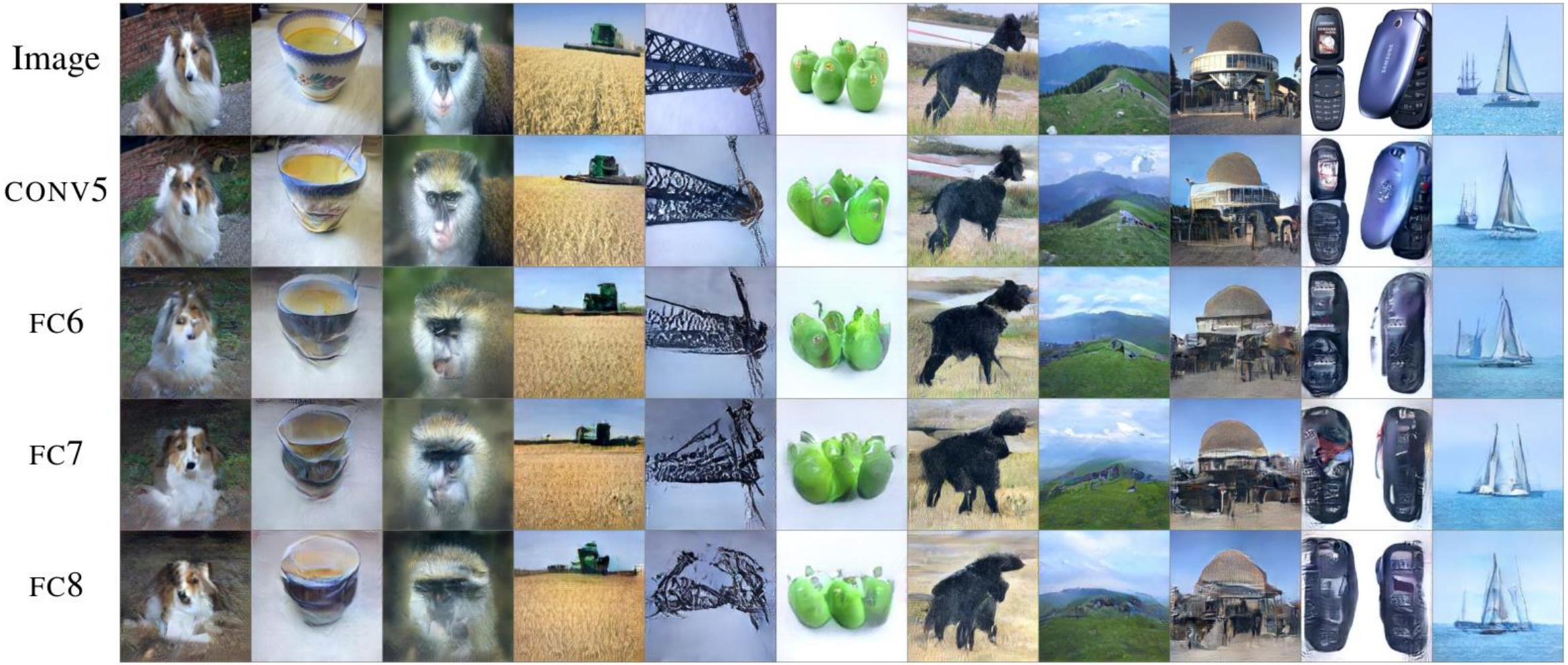
Results

- **Comparison** with other algorithms



Results

- Reconstructions from **CONV and FC**



Results

- Interesting reconstruction results by **interpolating** two feature representations.

Discussion

(Bellows are Personal Comments, not mentioned in the paper. Maybe some mistakes.)

- Image reconstruction based on **image iteration VS model iteration** (this paper)
  - Image iteration based method is slow, but can be applied to reconstruct from any layer **without pre-training**.
  - Model iteration based method is fast, but needs to **pre-train a layer-specific model**.
- For the layer with the same input and output dimension, can the model **trained to reconstruct the layer in AlexNet** be applied to **VGG**?
  - If not, is it possible to propose a method to make the trained model more **extensive**?

Synthesizing the preferred inputs for neurons in neural networks via deep generator networks

NIPS 2016

Anh Nguyen, Alexey Dosovitskiy, Jason Yosinski, Thomas Brox and Jeff Clune

Introduction

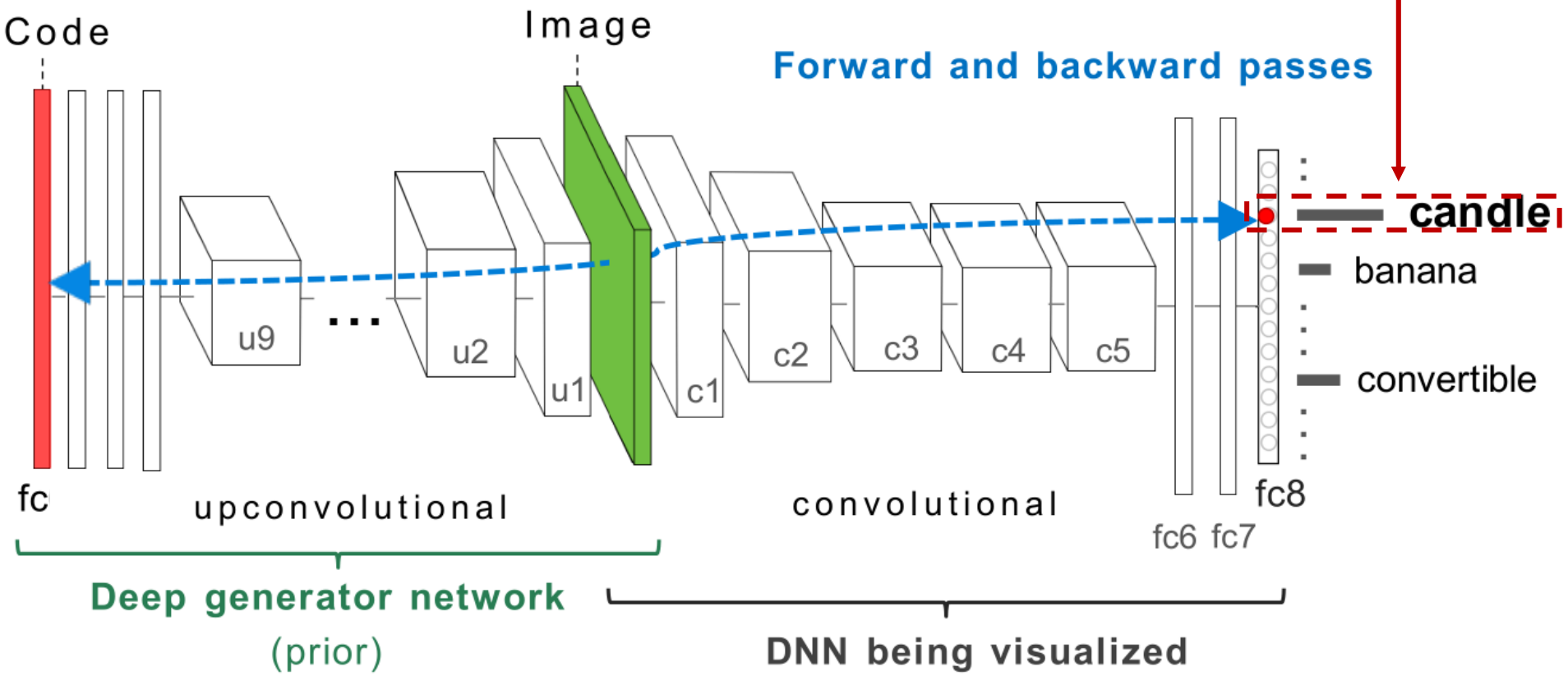
- Title: Synthesizing the preferred inputs for neurons
- Contributions of this paper:
  - Generates qualitatively synthetic images that look almost real **for a specific class**.
  - Reveals the features learned by each **neuron**. (neuron-level)

Real

Generated

Proposed Method

- Assume we want to synthesize the preferred input for “candle” class output neuron.
  (i.e., produce candle natural images) **(Two conditions: natural image, highly activate “candle” neuron)**



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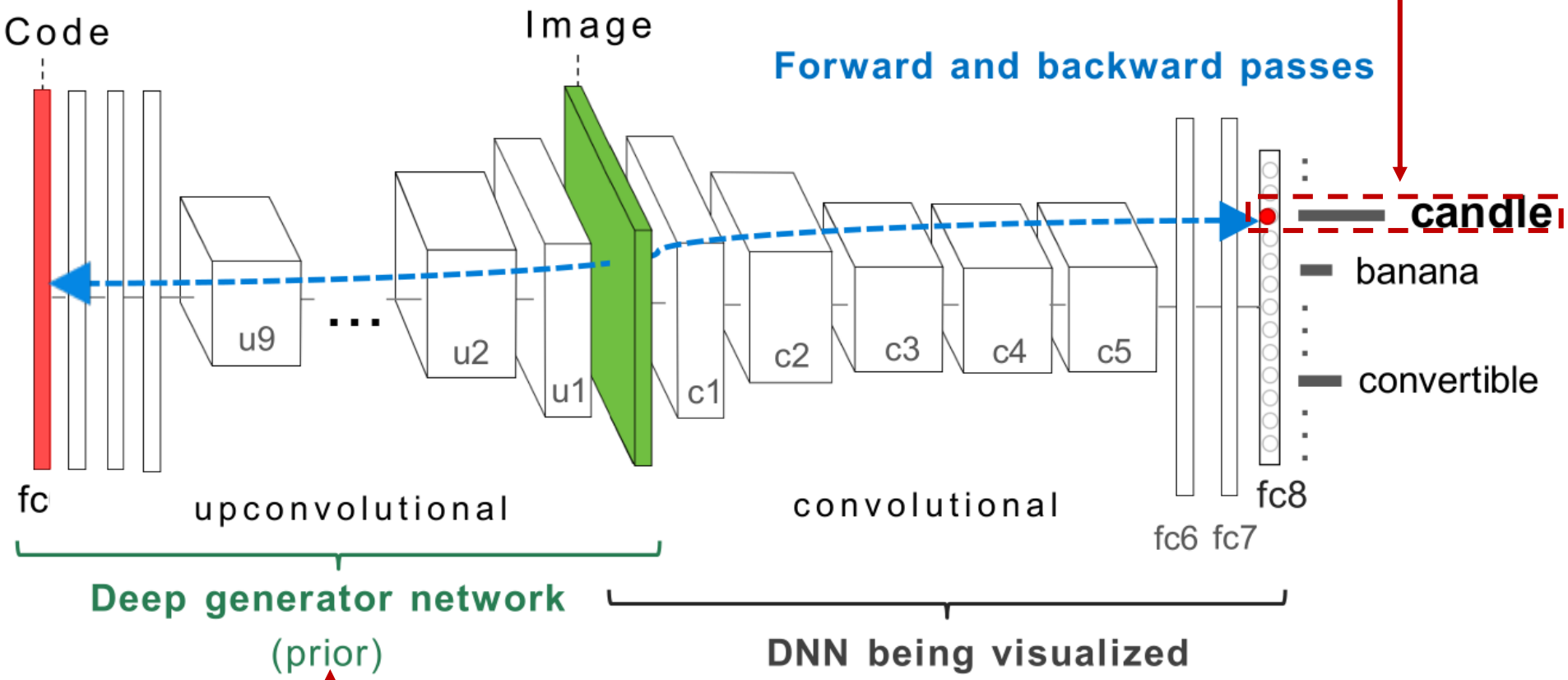

Fixed. Use **trained model** in my **previous presented paper**. The motivation is that as trained model is capable of producing natural images, it can be used as a **natural image prior**.

Proposed Method

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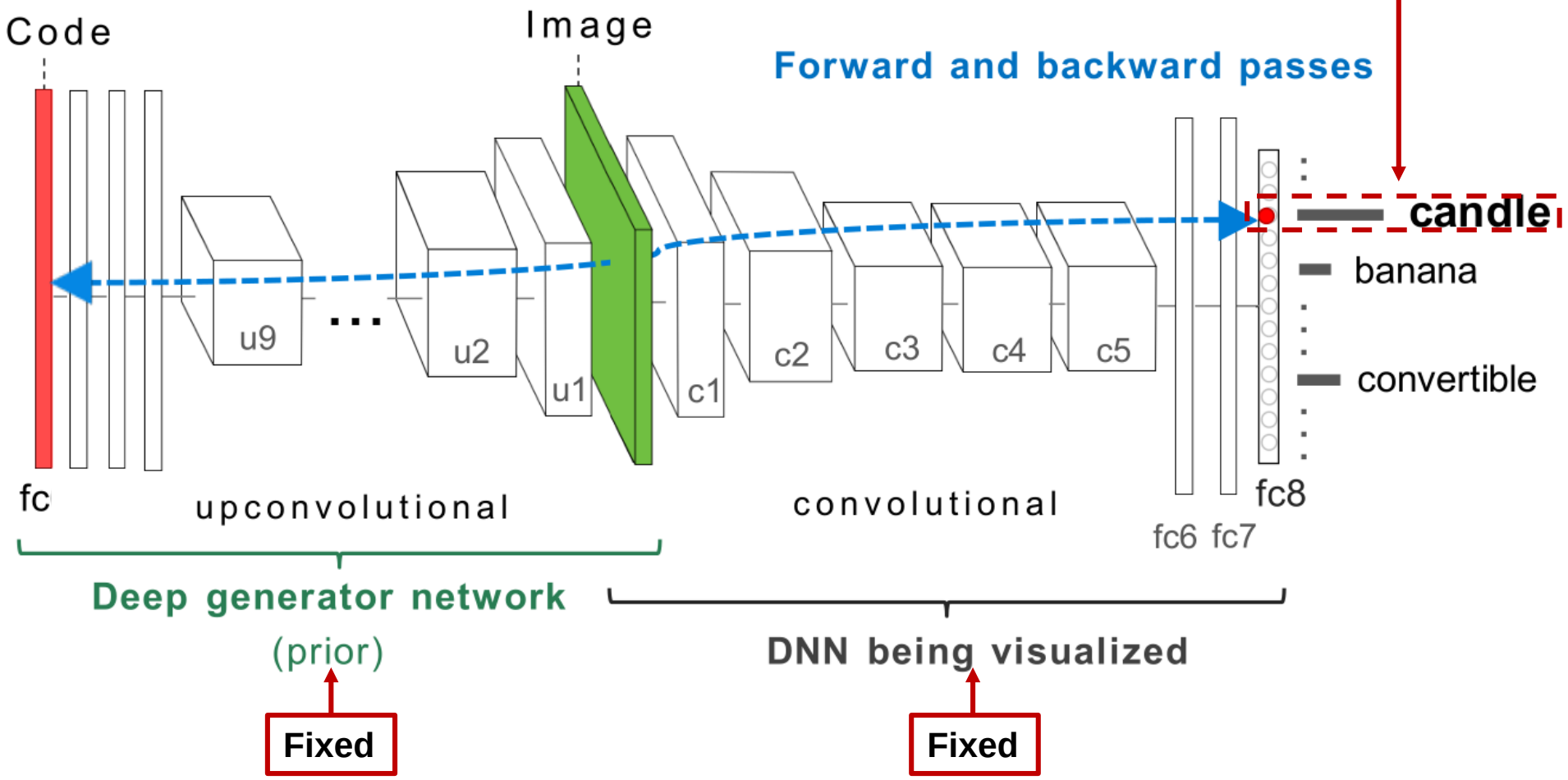

Still layer-specific trained model. One layer, one model

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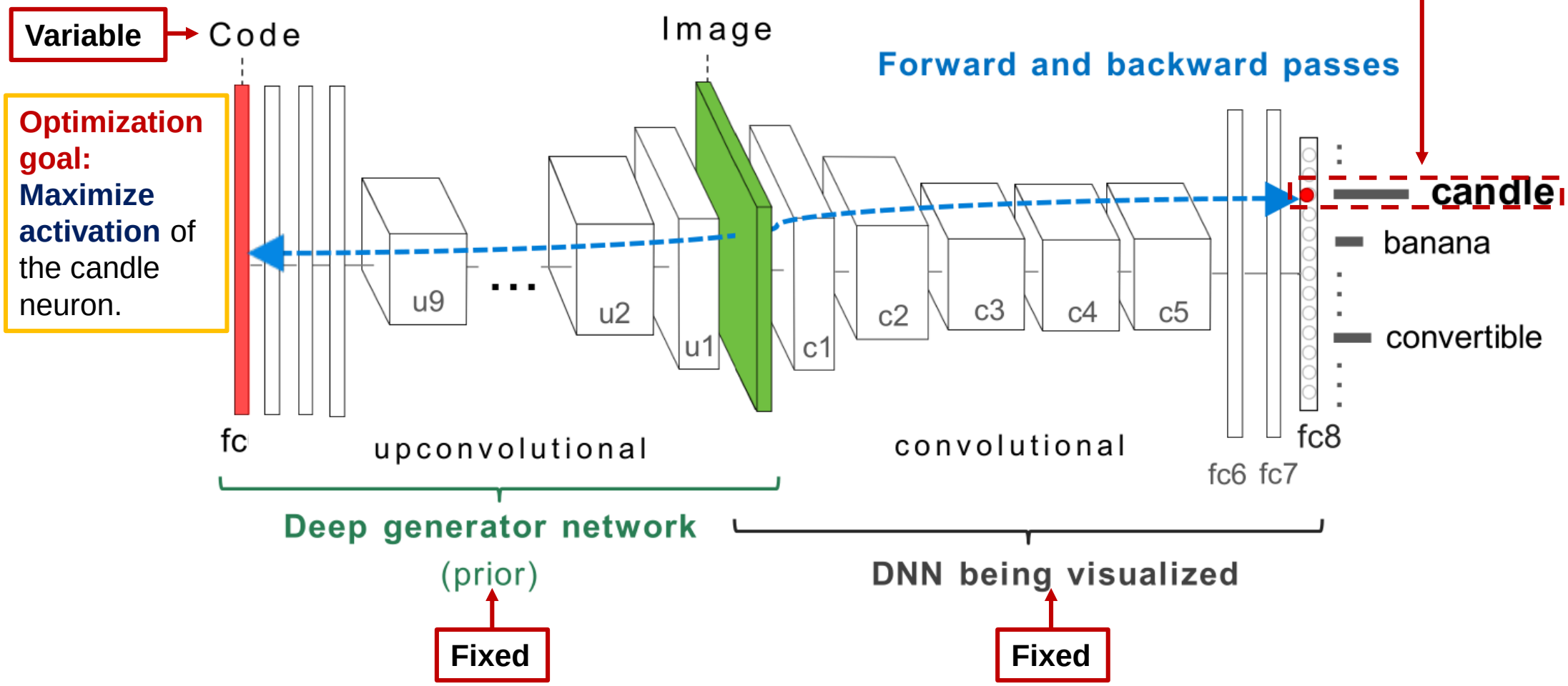

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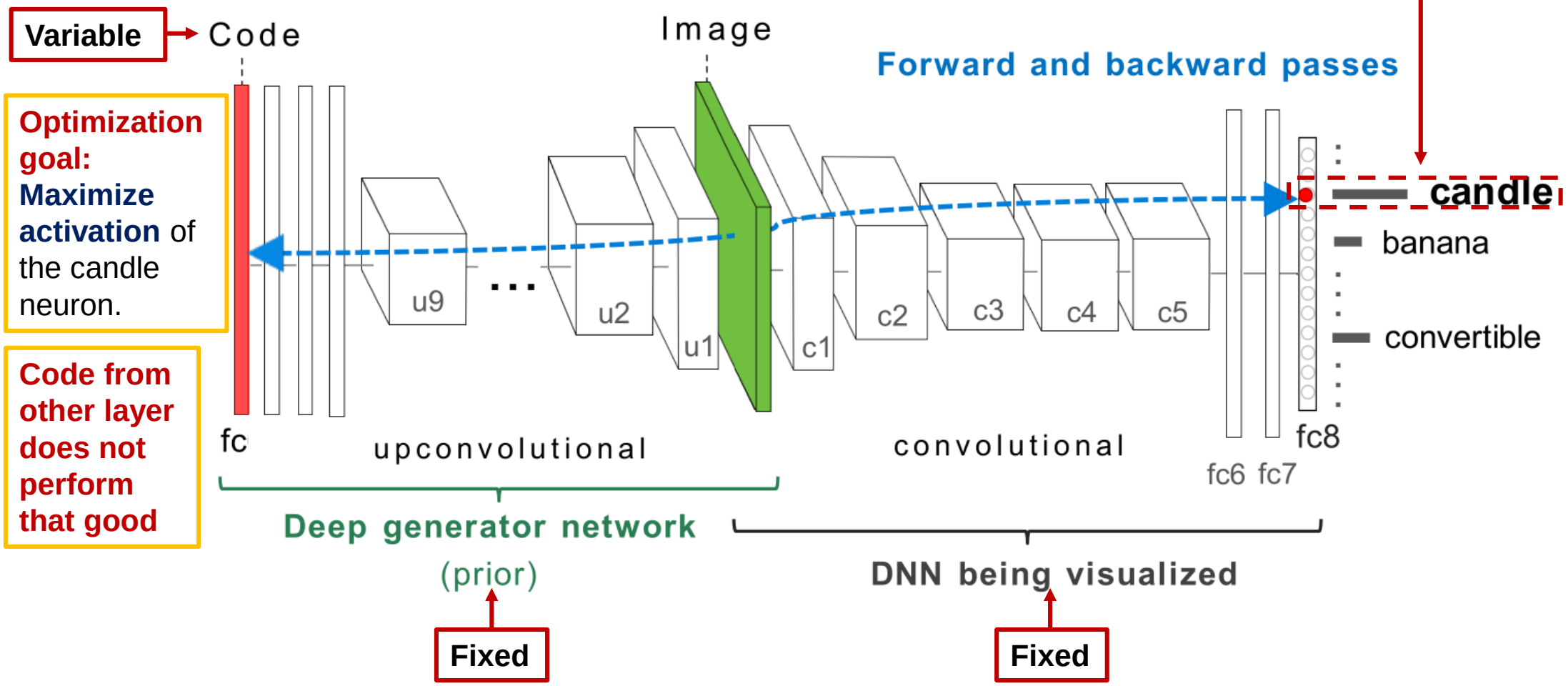

Still layer-specific trained model. One layer, one model

Results

- **Comparison** with other algorithms:

GT

This paper

Discussion

- Problem existing in the proposed algorithm (DGN-AM): ***Diversity***

9 images that **high activate** the neuron

(a) Real: top 9

Discussion

- Problem existing in the proposed algorithm (DGN-AM): ***Diversity***

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(a) Real: top 9 (b) DGN-AM

Discussion

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This paper

(a) Real: top 9 (b) DGN-AM (c) Real: random 9

Discussion

- Problem existing in the proposed algorithm (DGN-AM): ***Diversity***

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This paper

CVPR 2017 spotlight

Personal Comments and Ideas

(Bellows are Personal Comments, not mentioned in the paper. Maybe some mistakes.)

- **Comparison of my presented two papers:**

Paper #1	Paper #2
Layer level. Construct what layer representation learns.	**Neuron** level. Construct what neuron representation learns.
Optimize **network**	Optimize **code** which is the input of network
-	Depend on the Generator trained using #1 paper's algorithm

Personal Comments and Ideas

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- **Future Potential Applications** of Image Reconstruction:
  - **Data Augmentation**
    - Design **task-specific image prior**
  - Anther way of **"Image Compression"**
    - **NOT** the general image compression.
    - **Save images** in the form of representation and use **pre-trained G** to produce the actual image.
    - Need to study how to completely reconstruct the same image as the original one

Thanks!